

A disproof of Conjecture 00000000753

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Counterexample at index zero

The conjecture proposes the exact finite count

$$p^{i(i+1)/2}$$

for polynomials whose coefficients satisfy $|a_i|_p \leq p^{-i}$ termwise.

Theorem 1. *The formula fails already at $i = 0$.*

Proof. At $i = 0$ the condition is $|a_0|_p \leq 1$. Both constant polynomials $P_0(X) = 0$ and $P_1(X) = 1$ satisfy it, since $|0|_p = 0$ and $|1|_p = 1$. They are distinct, so there are at least two admissible polynomials. The displayed formula gives

$$p^{0(0+1)/2} = p^0 = 1,$$

a contradiction. □

Moreover, if coefficients range in $\mathbb{Q}_p$ as the statement suggests, the unit ball supplies infinitely many constant polynomials. A finite count would require an unstated quotient or finite coefficient alphabet.

Lean 4 certificate

The project in `lean4/` represents constant polynomials by their integer coefficient, records the exact index-zero integrality condition, constructs 0 and 1, proves they are distinct, and evaluates the asserted formula to 1. No `sorry`, `native_decide`, or added axiom is present.